

Chapter 1: DISCRETE-TIME SIGNALS AND SYSTEMS

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Prob 1:

Determine whether the system is linear and time-invariant:

- a. $y[n] = 3x[n] + 5$
- b. $y[n] = x^2[n - 1] + x[2n]$
- c. $y[n] = x[n^3]$
- d. $y[n] = nx[n - 3] + 3x[n]$
- e. $y[n] = n + 3x[n]$

Prob 2:

Given the difference equation of the LTI system, determine the causal impulse response $h[n]$:

- a. $y[n] = 3x[n] - 2x[n - 1] + 4x[n - 3]$
- b. $y[n] = 4x[n] + x[n - 1] - 3x[n - 3]$
- c. $y[n] = x[n] - 3x[n - 3]$
- d. $y[n] = 0.9y[n - 1] + x[n]$
- e. $y[n] = 0.64y[n - 2] + x[n]$
- f. $y[n] = 0.5y[n - 1] + 4x[n] + x[n - 1]$

Prob 3:

Given the impulse response of the LTI systems, determine the difference equation:

- a. $h[n] = (0.9)^n u[n]$
- b. $h[n] = (-0.6)^n u[n]$
- c. $h[n] = (0.9)^n u[n] + (-0.9)^n u[n]$
- d. $h[n] = (0.9j)^n u[n] + (-0.9j)^n u[n]$

Prob 4:

The LTI system has

$$h[n] = \left(\frac{2}{3}\right)^n (u[n] - u[n-5]) :$$

- a. Plot $h[n]$;
- b. Determine whether the system is stable, causal, linear, time invariant, and memoryless;
- c. Determine the output if the input is $x[n] = 2u[n] - u[n-1] - u[n-2]$;
- d. Write the difference equation of the system.